

Pablo Arango

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EDUCATION

The University of Texas at Austin, Austin, TX (Aug 2024 - Dec 2027)

Bachelor of Computer Science (Honors) and Mathematics

Honors: Turing Scholars Program Member

Relevant Coursework: Discrete Mathematics Honors, Data Structures Honors, Computer Organization and Architecture Honors, Algorithms and Complexity Honors, Computer Systems Honors, Linear Algebra, Game Theory, Number Theory, Wireless Network, Generative Visual Computing

EXPERIENCE

Illuma Alliance, Durham, North Carolina

Data Science & Software Development Intern — May 2026 to August 2026

- Architected and deployed a custom Model Context Protocol (MCP) using **Python**, **LangChain**, and **Streamlit** to analyze, track, and extract actionable trends from complex IoT farm sensor data.
- Engineered a real-time computer vision pipeline** utilizing YOLO, Roboflow, and OpenCV to identify livestock and detect distinct physical features from live camera feeds.

Asimetrix, Medellin, Colombia

Data Science & Software Development Intern — May 2025 to July 2025

- Designed and deployed an AI-driven retrieval and analytics agent for large-scale private datasets using **Model Context Protocol (MCP)**.
- Built Retrieval-Augmented Generation (RAG) pipelines using **AWS Bedrock**, **LangChain**, **Agno**, **Chroma**, **Pinecone**, and custom knowledge bases.
- Enabled automated trend detection, semantic search, and high-accuracy query answering across high-volume enterprise data.

Somnium – University of Navarra / The Columbus School, Medellín, Colombia & Madrid, Spain

Researcher — Nov 2023 to Feb 2024

- Developed an Arduino-based chemical release system with embedded control logic to deliver compounds at targeted physiological sleep phases.
- Optimized system timing, reliability, and repeatability for human sleep-cycle experimentation.

Landers CIA, Medellín, Colombia

Technology Intern — May 2023 to Jul 2023

- Conducted company-wide technology rotation across industrial and managerial departments.
- Diagnosed data-management and process inefficiencies and produced recommendations to improve operational workflows.

PROJECTS

- Compiler** (January - February 2025)
 - Developed a **high-performance compiler** in C for a custom-designed language name Fun, implementing all basic functionalities including lexical analysis, parsing, semantic analysis, and code generation.
- Parallel Distribution System** (April - May 2025)
 - Developed and integrated a custom compression algorithm into a parallel distributed system, achieving a **70% reduction in data size** and surpassing initial performance goals by significantly boosting throughput and demonstrating proficiency in concurrency and inter-process communication.
- Multithreaded Coroutines Library** (April - May 2025)
 - Architected and implemented a **multithreaded coroutine library** to enable highly efficient, cooperative multitasking and asynchronous operations in a user-space environment. | Managed over 7.5M Coroutines
- Multicycled Pipelined CPU** (April 2025)
 - Architected and implemented a multicycle, **five-stage pipelined CPU** entirely in **Verilog**. Implemented comprehensive **data and control hazard handling** using forwarding and stalling logic to maximize pipeline utilization and minimize idle cycles.
- Kernel** (August - October 2025)
 - Engineered core OS features in C and Assembly, including **virtual memory** management (paging and segmentation), a **multithreaded scheduler** for process parallelism, efficient file systems, and an **interrupt handling system**. Implemented on **x86-64 architecture**, this kernel runs a multilevel feedback queue.

LEADERSHIP & COMMUNITY INVOLVEMENT

Tikkun Olam Makers, Latin America (Aug 2019 - June 2024)

Regional Leader

- Led a non-profit creating an inclusive society for people with disabilities. Directed development of tech solutions for 20+ individuals using industrial machines, design processes, and professional software, driving impactful accessibility innovations.

SKILLS

Technical/Computer Skills: JUnit, VSCode, Eclipse, Java, C, C++, Python, TypeScript, Verilog, Compiler building and design, Agno Framework, AWS Bedrock, Chroma/Pinecone, MCP, RAG, Database Management, PostgreSQL, MongoDB, SQL, Amazon Athena/S3

Languages: Advanced English and Spanish, Proficient Portuguese and French